

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CL)

December 2011

CL 305 Environmental Engineering - II

Date: 16.12.2011, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Give classification of wastes based on their nature. [05]
(b) Explain daily variation of sewage flow. [02]
- Q - 2 (a) Enlist different physical characteristics of Sewage and discuss their environmental significance. [07]
(b) Define attached growth process for sewage treatment. Explain various stages of bacterial growth with neat sketch. [05]
(c) Write significance of grit chamber in sewage treatment plant. [02]

OR

- Q - 2 (a) With the concept of first stage B.O.D. show that ultimate first stage B.O.D is equivalent to the organic matter present. [07]
(b) Explain Chemical Unit Processes for sewage treatment. [05]
(c) For a domestic sewage having 5 day B.O.D. of 200 mg/l @20 °C, calculate the B.O.D. at 1 day @ 12°C. Assume $k_D = 0.18$ @ 20 °C. [02]
- Q - 3 (a) Design screen chamber of sewage flow of 32 million liters per day. [05]
(b) Design a trickling filter to treat 20 million liters per day sewage of BOD 210 mg/l. The final effluent should be of 30 mg/l B.O.D. [05]
(c) Describe in brief various types of settling. [04]

OR

- Q - 3 (a) Design primary settling tank for the flow 40 m³/hr. Take surface loading rate 40m³/day and incoming SS in PST = 470 mg/L. [05]
(b) Differentiate between B.O.D. and C.O.D. Why is C.O.D. of sewage sample always higher than B.O.D? [05]
(c) Draw the process flow diagram for the treatment of sewage. How sewage discharge can be calculated? [04]

SECTION – II

- Q - 4 (a) Explain methods of solid waste collection. [05]
(b) Describe composition of solid waste. [02]
- Q - 5 (a) Define air pollution. Describe adsorption and condensation for gaseous pollution control. [07]
(b) Explain incineration and landfilling of Solid Waste. [05]
(c) Enlist preventive measures for air pollution control. [02]

OR

- Q - 5 (a) Give classification of air pollutants. Enlist advantages and drawbacks of electrostatic precipitator. [07]
(b) Explain composting of solid waste. [05]
(c) Enlist control measures for particulate matter control. [02]
- Q - 6 (a) Classify the sources of air pollution based on their origin. Enlist effects on property due to air pollution. [05]
(b) Explain principle and working of cyclone. [05]
(c) Define environmental audit. Enlist the members involved in environmental audit. [04]

OR

- Q - 6 (a) Describe how air pollution from automobiles exhaust can be controlled. [05]
(b) Write a note on: gravity settling chamber. [05]
(c) Explain the need of environmental audit. [04]
